



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-2

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Subject Name: System Design

Subject Code: 23CSH-314

1. **AIM :** To design an online E-commerce platform similar to Amazon/Flipkart for browsing and purchasing products like mobiles, laptops, cameras, and clothes.
To implement Kafka, Elasticsearch, and a CDC pipeline for real-time data processing, fast search, and scalability.

2. Objective:

- To develop a scalable online shopping system for product listing, search, and order management.
- To use Apache Kafka for real-time event streaming and inter-service communication.
- To implement Elasticsearch for fast and efficient product search.
- To integrate a CDC pipeline for real-time synchronization between databases and services.

3. Tools Required:

- Programming Language: Java / Python / JavaScript
- Backend Framework: Spring Boot / Express.js / Flask
- Database: MySQL / PostgreSQL / MongoDB
- API Testing Tool: Postman
- Design Tool: Draw.io (for HLD diagrams)
- Web Browser
- ElasticSearch
- Kafka
- CDC connector

4. SYSTEM DESIGN / SYSTEM SPECIFICATION:

4.1. Functional Requirements:

- User should be able to search and find the products based on product title or names.
- User should be able to view the details of the product like description, image, available quantity, review, accessed.
- User should be able to select the quantity and move the product/item into the cart.
- User should be able to make the payment and should be able to perform the check out.
- User should be able to check the status of the order.
- System should be able to manage purchase of items having limited stocks.

4.2. Non-functional Requirements:

- Target Scale: 100 Million DAU with 10 orders processed per second. Availability – System should be available 24/7
- Consistency & Availability: Here for this system we need both as per the Target Scale. Now we should specify that, which part of our system needs what?
- Latency: Required: ~200 ms
- Scaling: Horizontal / Vertical Consistency

4.3. Core-Entities of the System:

1. User / Client
2. Product
3. Cart
4. Orders
5. Checkout followed by Payment

4.4. API Endpoints Creation:

1. GET API Call: Prod_Search

Https://Local_Host/products/search_item = {Search_keywords}

HTTP Req

```
{  
  GET: <iPhone 16>  
}
```

HTTP Res

```
{  
  <ProductID:iPhone>  
}
```

Now, on front-end if multiple data of respective product is coming in that case the FE becomes faulty -> ultimately increasing the LATENCY.

For that: we can use Pagination 1, 2, 3, 4, ----- SO ON

2. GET API Call: View Product Details

Https://Local_Host/products/{product_id}

HTTP Req

```
{  
  GET: <Product_id = 17>  
}
```

HTTP Res

```
{
```



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```
Product_id:17,  
Name: iPhone 17,  
Color: Navy Blue,  
Price: $1099,  
Image Thumbnail: URL_Image  
}
```

3. POST API Call: Item add in cart

Https://Local_Host/cart/add_products

```
HTTP Req  
{  
  Product_id = 17,  
  Product_id = 16  
}
```

```
HTTP Req Header  
{  
  User_id: 04  
}
```

```
HTTP Res  
{  
  Cart_id: 101  
}
```

4. PUT API Call: To update any order in the cart

Let's Suppose you want to add one more product into the cart.

5. DELETE API Call: To remove any item from the cart

Let's Suppose you want to delete one more product into the cart.

6. POST API Call: for check out & Payment Htps://Local_Host/checkout

-> {post body}

```
HTTP REQ  
{  
  All products ID's,  
  Total Quantity,  
  Total Price  
}  
HTTP RES  
{
```



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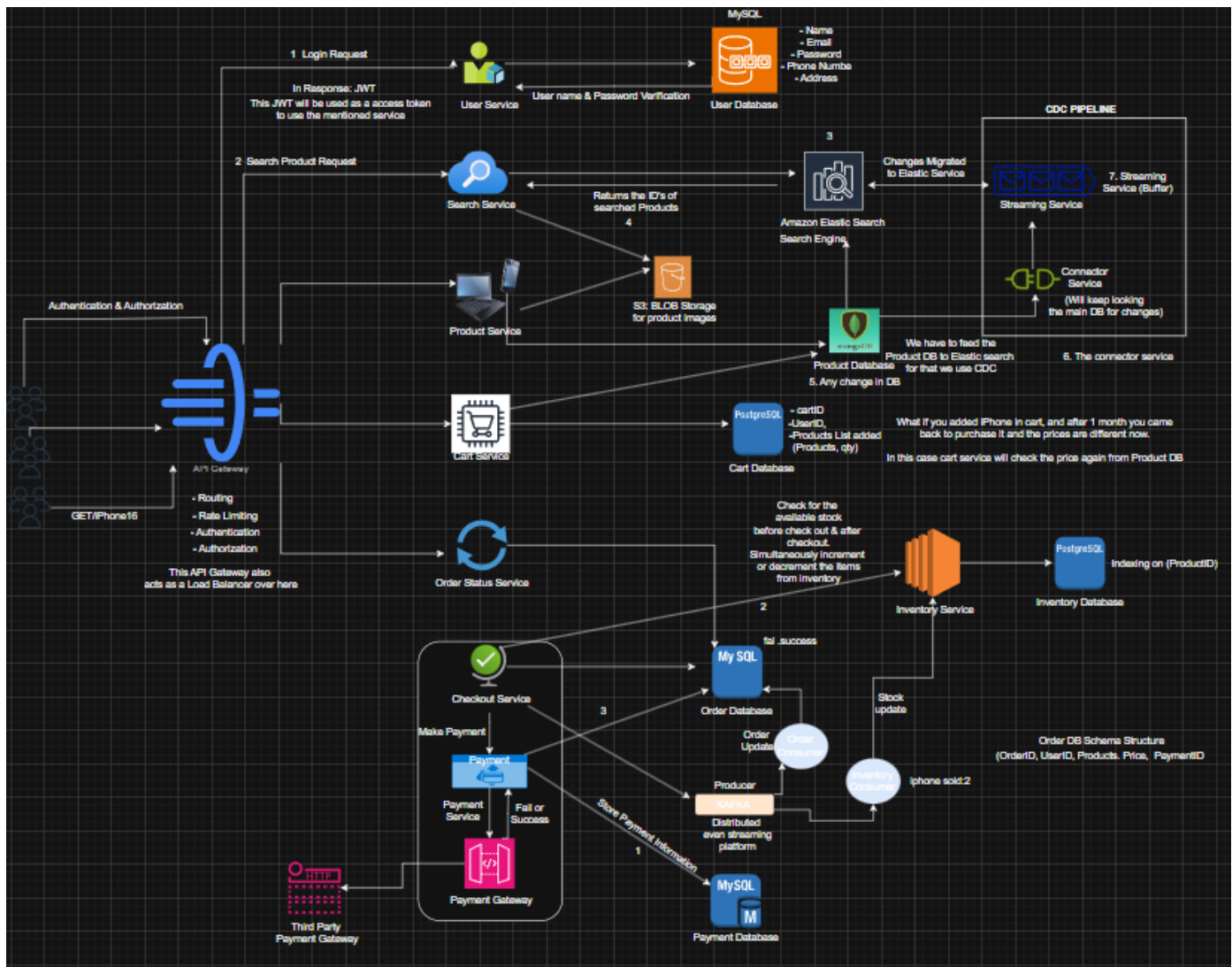
```
        Order_ID  
    }
```

```
Https://Local_Host/payment -> {post body}  
HTTP REQ  
{  
    Order ID,  
    Payment Type,  
    Payment Mode  
}  
HTTP RES  
{  
    Confirmation_Status: Succes / Fail  
}
```

7. GET API Call: Order Status

```
Https://Local_Host/orde_status = {order_id}
```

5. HLD(High Level Design



7. Learning Outcomes

- Understand the end-to-end architecture of a scalable e-commerce platform for product browsing and online purchases.
- Gain hands-on experience integrating **Apache Kafka** for real-time event streaming and data pipelines.
- Implement **Elasticsearch** to enable fast, full-text product search and efficient filtering.
- Design and deploy a **CDC (Change Data Capture) pipeline** for real-time synchronization between databases and search indexes.
- Learn to build a fault-tolerant, scalable system capable of handling high traffic and real-time data updates.